# CSE 270 Sports Analytics

Homework 2

Name: _____

## Poisson Regression

### Problem 1(20 points)

**Predicting football games using Poisson regression.**

1. Make predictions for the same game from the same league assigned to you in homework 1, but now use the poisson regression model instead of distributions.**(8 points)**

```
##          HW       D      ATW
## 1 2.501876 3.90625 2.904444
```

```
data("f_data_sm")

get_poisson_form <- function(data,country){
  country_data <- data %>%
    filter(COUNTRY == country)

  country_data_home <- country_data %>%
    group_by(HOMETEAM) %>%
    mutate(TEAM = HOMETEAM, OPPONENT = AWAYTEAM, GOAL = FTHG, HOME = 1) %>%
    ungroup() %>%
    select(TEAM, OPPONENT, GOAL, HOME)

  country_data_away <- country_data %>%
    group_by(AWAYTEAM) %>%
    mutate(TEAM = AWAYTEAM, OPPONENT = HOMETEAM, GOAL = FTAG, HOME = 0) %>%
    ungroup() %>%
    select(TEAM, OPPONENT, GOAL, HOME)

  return(rbind(country_data_home, country_data_away))
}


england_2 <- get_poisson_form(f_data_sm,"England")

model <- glm(GOAL~TEAM+OPPONENT+HOME, data=england_2, family=poisson(link=log))

everton <- data.frame(TEAM = "Everton", OPPONENT = "Man United", HOME = 1)
mu <- data.frame(TEAM = "Man United", OPPONENT = "Everton", HOME = 0)
```

```
(pred_everton <- predict(model,newdata=everton,type="response"))
```

```
##        1
## 1.058826
```

```
(pred_mu <- predict(model, newdata=mu, type="response"))
```

```
##       1
## 1.55874
```

2. Interpret the result and compare with the betting odds from homework 1.
   Is the same team expected to win ? **(2 points)**

The poisson regression model did better, as in reality Man United won 2-1.
The skellam distribution considered Everton as the favorite to win the game.

3. Define a metric to calculate the efficiency of the predictions. **(5 points)**

Let's divide the dataset into test and train sets.

```
train <- f_data_sm %>% filter(SEASON!=max(SEASON))

train_poisson <- get_poisson_form(train, "England")

model <- glm(GOAL~TEAM+OPPONENT+HOME, data=train_poisson, family=poisson(link=log))

test_poisson <-get_poisson_form(f_data_sm %>% filter(SEASON == max(SEASON)), "England")

test_poisson <- test_poisson %>%
  filter(TEAM %in% unique(train_poisson$TEAM),
         OPPONENT %in% unique(train_poisson$OPPONENT))

predicted <- predict(model,newdata=test_poisson,type="response")

test_poisson$P.G <- predicted

test_posson <- test_poisson %>%
  mutate(DIFF = GOAL - P.G)

sqrt(mean((test_poisson$GOAL - test_poisson$P.G)^2))
```

```
## [1] 1.250399
```

4. Find the under estimated and over estimated teams based on Poission model. **(5 points)**

```
(test_poisson %>%
  filter(GOAL > P.G))
```

```
## # A tibble: 315 x 5
##    TEAM        OPPONENT        GOAL  HOME   P.G
##    <chr>       <chr>          <dbl> <dbl> <dbl>
##  1 Man United  Leeds              5     1 1.97
##  2 Chelsea     Crystal Palace     3     1 2.15
##  3 Everton     Southampton        3     1 1.59
##  4 Watford     Aston Villa        3     1 1.16
##  5 Newcastle   West Ham           2     1 1.66
##  6 Aston Villa Newcastle          2     1 1.42
##  7 Leeds       Everton            2     1 1.53
##  8 Man City    Norwich            5     1 2.41
##  9 Brighton    Watford            2     1 1.38
## 10 Southampton Man United         1     1 0.983
## # ... with 305 more rows
## # i Use 'print(n = ...)' to see more rows
```

```
(test_poisson %>%
  filter(GOAL < P.G))
```

```
## # A tibble: 369 x 5
##    TEAM      OPPONENT   GOAL  HOME   P.G
##    <chr>     <chr>     <dbl> <dbl> <dbl>
##  1 Burnley   Brighton      1     1 1.21
##  2 Leicester Wolves        1     1 1.79
##  3 Norwich   Liverpool     0     1 0.923
##  4 Tottenham Man City      1     1 1.48
##  5 Liverpool Burnley       2     1 2.25
##  6 Wolves    Tottenham     0     1 1.18
##  7 Arsenal   Chelsea       0     1 1.50
##  8 Brighton  Everton       0     1 1.05
##  9 Norwich   Leicester     1     1 1.29
## 10 Liverpool Chelsea       1     1 1.46
## # ... with 359 more rows
## # i Use 'print(n = ...)' to see more rows
```

# Bradley terry model

## Problem 2(30 points)

**Construct Bradley-Terry model for NBA regular season games.**

1. Load the dataset **nba_2009_2021** from the package **SportsAnalytics270** and remove the game Boston Celtics vs Indiana Pacers by filtering the **home.PTS** and **away.PTS** variables to be equal to 0.**(3 points)**

```
nba2009_2021 <- nba2009_2021 %>%
  filter(home.PTS != 0, away.PTS != 0)
```

2. Prepare the dataset for fitting a Bradley-Terry model by adding 2 variables, **ht_w** and **at_w**. Assign the values of the varibles to (**ht_w** = 1, **at_w** = 0) if the home team won the game and (**ht_w** = 0, **at_w** = 1) if the away team won the game. **(5 points)**

```
nba <- nba2009_2021 %>% select(SEASON_ID, home.TEAM_NAME, away.TEAM_NAME, home.WL) %>%
          mutate(ht_w = ifelse(home.WL=="W",1,0),
                 at_w = ifelse(home.WL=="L",1,0),
                 home.TEAM_NAME = as.factor(home.TEAM_NAME),
                 away.TEAM_NAME = as.factor(away.TEAM_NAME))

train_nba <- nba %>% filter(SEASON_ID!="2021")  %>%
        group_by(home.TEAM_NAME, away.TEAM_NAME) %>%
        summarise(ht=sum(ht_w), at=sum(at_w))

test_nba <- nba %>% filter(SEASON_ID=="2021")  %>%
        group_by(home.TEAM_NAME, away.TEAM_NAME) %>%
        summarise(ht=sum(ht_w), at=sum(at_w))
```

3. Convert the variables representing the home team and the away team into a factor. **(2 points)**

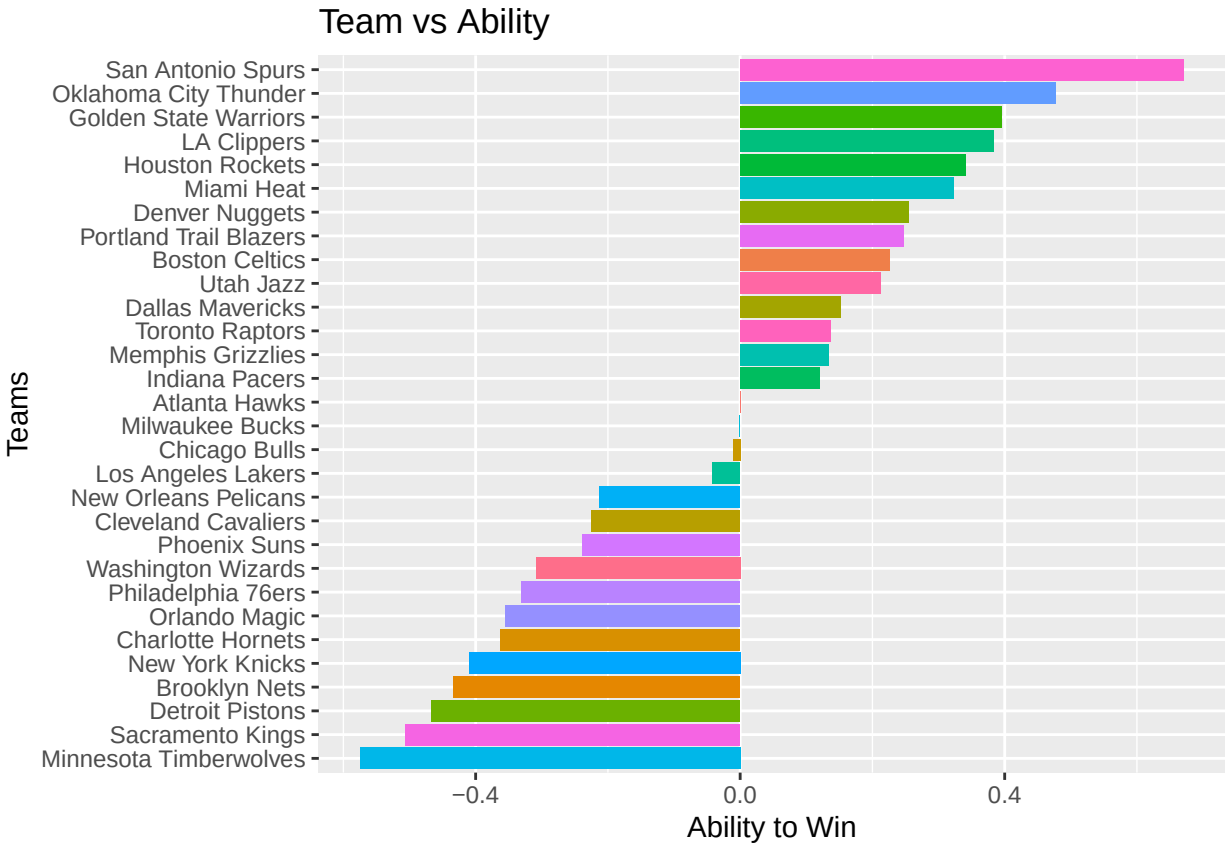
4. Fit a Bradley-Terry model into the dataset to get the abilities of the teams to win. **(5 points)**

```
BT_Model <- BTm(cbind(ht,at), home.TEAM_NAME, away.TEAM_NAME, data=train_nba, id="team_")
```

5. Plot the abilities of the teams. **(5 points)**

```
abilities <- as.data.frame(BTabilities(BT_Model))
abilities$team <- rownames(abilities)
abilities <- abilities[order(abilities$ability, decreasing = T),]

ggplot(data=abilities, aes(x=reorder(team,ability), y = ability, fill =team))+ geom_bar(stat="identity"
  labs(x="Teams", y="Ability to Win", title="Team vs Ability")
```




6. Interpret the results of the plot, which team is the best and which team is the worst ? Why is the ability to win for team **Atlanta Hawks** equal to 0 ? **(5 points)**

Minnesota the worst, Spurs the best, all the abilities calculated relative to Atlanta Hawks.

7. Make a prediction for the upcoming 3 games.
(Schedule can be found here: https://www.nba.com/schedule ) **(5 points)**

```
pred<-predict(BT_Model,newdata = test_nba, type='response')

test_nba$home_win<-pred
test_nba$away_win<-1-pred
head(test_nba)
```

```
## # A tibble: 6 x 6
## # Groups:   home.TEAM_NAME [1]
##   home.TEAM_NAME away.TEAM_NAME         ht    at home_win away_win
##   <fct>          <fct>               <dbl> <dbl>    <dbl>    <dbl>
## 1 Atlanta Hawks  Boston Celtics          2     0    0.444    0.556
## 2 Atlanta Hawks  Brooklyn Nets           1     1    0.607    0.393
## 3 Atlanta Hawks  Charlotte Hornets       1     1    0.590    0.410
## 4 Atlanta Hawks  Chicago Bulls           1     1    0.503    0.497
## 5 Atlanta Hawks  Cleveland Cavaliers     2     0    0.556    0.444
## 6 Atlanta Hawks  Dallas Mavericks        1     0    0.462    0.538
```

```
# Washington Wizards vs Charlotte Hornets
# Philadelphia 76ers vs Cleveland Cavaliers
# Houston Rockets vs Miami Heat

games <- data.frame(home.TEAM_NAME = c("Washington Wizards", "Philadelphia 76ers", "Houston Rockets"),
                    away.TEAM_NAME = c("Charlotte Hornets","Cleveland Cavaliers", "Miami Heat"))

home_relevel <- levels(test_nba$home.TEAM_NAME)
away_relevel <- levels(test_nba$away.TEAM_NAME)
games$home.TEAM_NAME <- factor(games$home.TEAM_NAME,
                           home_relevel)
games$away.TEAM_NAME <- factor(games$away.TEAM_NAME,
                           away_relevel)
games_prob <- predict(BT_Model,
                      newdata = games,
                      level = 2,
                      type = "response")
games_df <- data.frame(games,
                       ht_w = games_prob,
                       at_w = 1 - games_prob)

games_df
```

```
##       home.TEAM_NAME      away.TEAM_NAME      ht_w      at_w
## 1 Washington Wizards   Charlotte Hornets 0.5135425 0.4864575
## 2 Philadelphia 76ers Cleveland Cavaliers 0.4733957 0.5266043
## 3    Houston Rockets          Miami Heat 0.5045306 0.4954694
```

# Winning Percentage

## Pythagorean wins for european football

## Problem 3 (30 points)

1. Library **SportsAnalytics270** has a function final_table. It creates final league standing for the season. Use this function to create a dataframe with final standings of all seasons of your league. Combine seasons into 1 datafrmae. You need to get something like nba_east. **(10 points)**

```
final_tables <- function(data = f_data_sm,country){
  result <- c()
  for(season in unique(data$SEASON)){
    season_table <- final_table(data, country, season)
    league <- data[data$SEASON == season &
                     data$COUNTRY == country,]$LEAGUE[1]
    season_table$SEASON = season
    season_table$COUNTRY = country
    season_table$LEAGUE = league
    result <- rbind(result, season_table)
  }
  cols <- colnames(result)
  len <- length(cols)
```

```
  result <- result[c("SEASON","COUNTRY","LEAGUE",cols[1 : (len-3)])]
  return(as.data.frame(result))
}
england_tables <- final_tables(country = "England")
```

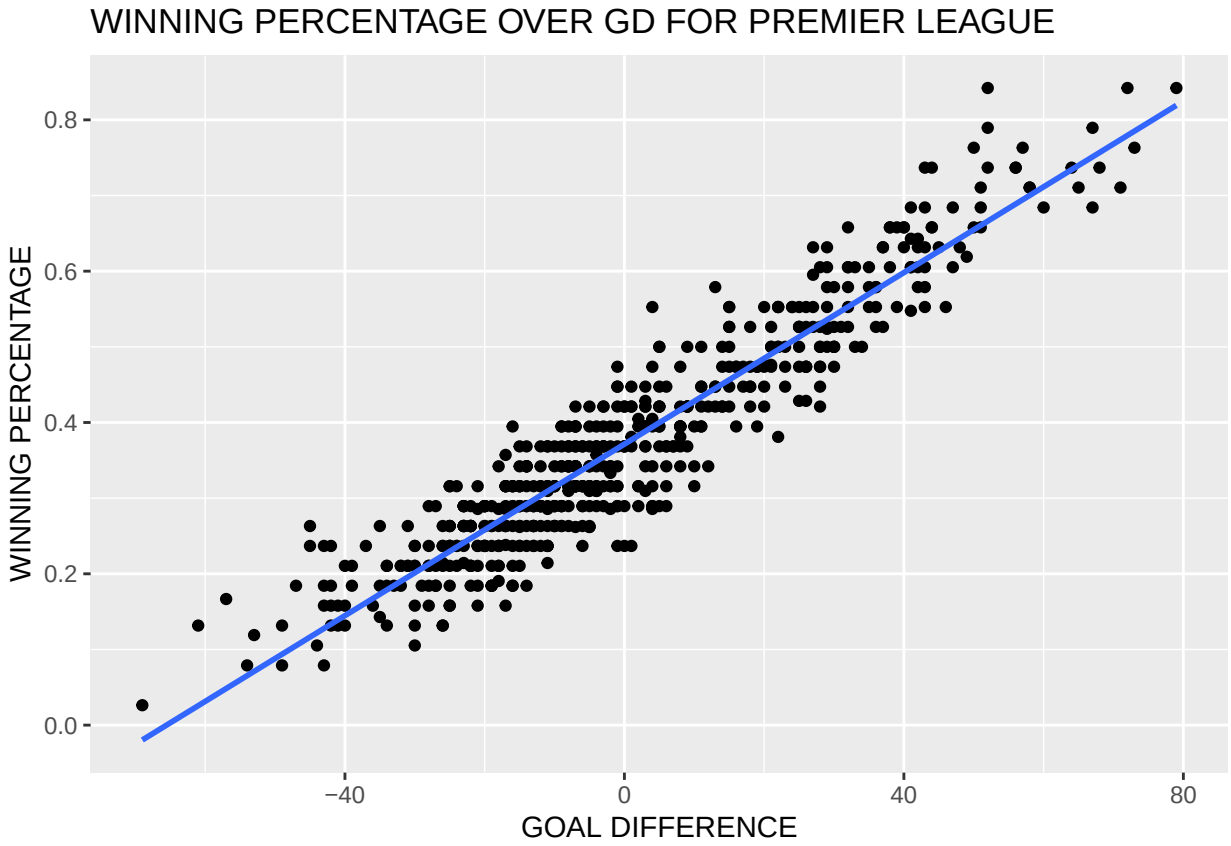
2. Create a variable for winning percentage and goal difference. Note, as there are draws in football, we are going to take draw as a half win. **(2 points)**

```
calculate_wpct <- function(data){
  result <- c()
  for(index in 1:nrow(data)){
    team <- data[index,]
    team$WPCT <- (team$W) / (team$M)
    team$WPCTv2 <- (team$W + team$D/2)/(team$M)
    team$WPCTv3 <- (team$W + round(team$D / 3)) / (team$M)
    result <- rbind(result,team)
  }
  return(as.data.frame(result))
}


england_tables$GD <- england_tables$GF - england_tables$GA
england_tables$WPCT <- england_tables$W / (england_tables$M)
```

3. Plot Goal differential against Winning Percentage with a regression line.
   Make sure to have a title and apropriate axis labels for your plot. **(5 points)**

```
england_tables %>%
  ggplot(aes(x = GD, y = WPCT)) +
  geom_point() +
  geom_smooth(method = "lm", se = F) +
  labs(x = "GOAL DIFFERENCE", y = "WINNING PERCENTAGE") +
  ggtitle("WINNING PERCENTAGE OVER GD FOR PREMIER LEAGUE")
```




4. Interpret the plot, do you think there is a strong correlation between the variables ? **(3 points)**

The relationship seems to be very strong, the teams over the lines are the ones who are overperforming, and the ones below the line are the underperforming teams.

5. Calculate Pearson correlation coefficient for Goal differential and Winning Percentage. **(2 points)**

```
cor.test(england_tables$GF, england_tables$WPCT)
```

```
##
##  Pearson's product-moment correlation
##
## data:  england_tables$GF and england_tables$WPCT
## t = 45.958, df = 582, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.8665405 0.9017763
## sample estimates:
##      cor
## 0.885424
```

6. Build a regression model to estimate the value for k from the Pythagorean formula **(3 points)**

```
england_tables_reg <- england_tables %>%
  filter(L != 0) %>%
  mutate(W = W + 0.5*D, L = L + 0.5*D, RATIO = GF / GA)

model <- lm(log(W/L)~0+log(RATIO), data=england_tables_reg)
summary(model)
```

```
##
## Call:
## lm(formula = log(W/L) ~ 0 + log(RATIO), data = england_tables_reg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50723 -0.09537  0.01826  0.11988  0.87642
##
## Coefficients:
##            Estimate Std. Error t value Pr(>|t|)
## log(RATIO)  1.19543    0.01396   85.66   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1674 on 582 degrees of freedom
## Multiple R-squared:  0.9265, Adjusted R-squared:  0.9264
## F-statistic:  7337 on 1 and 582 DF,  p-value: < 2.2e-16
```

5. Using the estimated value for k, calculate Pythagorean Winning percentage and Pythagorean wins for each team. **(3 points)**

```
k <- model$coefficients[1]
head(england_tables <- england_tables %>%
  mutate(PWPCT = ( GF ^ k / (GF ^ k + GA ^ k)), PW = M * PWPCT),5)
```

```
##   SEASON COUNTRY         LEAGUE       TEAM  M  W  D  L GF GA DIFF POINTS
## 1   1994 England Premier League Man United 42 27 11  4 80 38   42     92
## 2   1994 England Premier League  Blackburn 42 25  9  8 63 36   27     84
## 3   1994 England Premier League  Newcastle 42 23  8 11 82 41   41     77
## 4   1994 England Premier League    Arsenal 42 18 17  7 53 28   25     71
## 5   1994 England Premier League      Leeds 42 18 16  8 65 39   26     70
##   POSITION GD      WPCT     PWPCT       PW
## 1        1 42 0.6428571 0.7088743 29.77272
## 2        2 27 0.5952381 0.6612745 27.77353
## 3        3 41 0.5476190 0.6960601 29.23452
## 4        4 25 0.4285714 0.6819583 28.64225
## 5        5 26 0.4285714 0.6480900 27.21978
```

6. Find over performing and under performing teams. **(2 points)**

```
head(england_tables %>%
  filter(W > PW),5)
```

```
##   SEASON COUNTRY         LEAGUE       TEAM  M  W D  L  GF GA DIFF POINTS
```

```
## 1   2000 England Premier League Man United 38 28 7  3  97 45   52     91
## 2   2013 England Premier League Man United 38 28 5  5  86 43   43     89
## 3   2014 England Premier League  Tottenham 38 21 6 11  55 51    4     69
## 4   2017 England Premier League    Chelsea 38 30 3  5  85 33   52     93
## 5   2018 England Premier League   Man City 38 32 4  2 106 27   79    100
##   POSITION GD      WPCT     PWPCT       PW
## 1        1 52 0.7368421 0.7146639 27.15723
## 2        1 43 0.7368421 0.6960601 26.45028
## 3        6  4 0.5526316 0.5225506 19.85692
## 4        1 52 0.7894737 0.7560316 28.72920
## 5        1 79 0.8421053 0.8368352 31.79974
```

```
head(england_tables %>%
    filter(W < PW),5)
```

```
##   SEASON COUNTRY         LEAGUE       TEAM  M  W  D  L GF GA DIFF POINTS
## 1   1994 England Premier League Man United 42 27 11  4 80 38   42     92
## 2   1994 England Premier League  Blackburn 42 25  9  8 63 36   27     84
## 3   1994 England Premier League  Newcastle 42 23  8 11 82 41   41     77
## 4   1994 England Premier League    Arsenal 42 18 17  7 53 28   25     71
## 5   1994 England Premier League      Leeds 42 18 16  8 65 39   26     70
##   POSITION GD      WPCT     PWPCT       PW
## 1        1 42 0.6428571 0.7088743 29.77272
## 2        2 27 0.5952381 0.6612745 27.77353
## 3        3 41 0.5476190 0.6960601 29.23452
## 4        4 25 0.4285714 0.6819583 28.64225
## 5        5 26 0.4285714 0.6480900 27.21978
```

# Elo Ratings

## Problem 4 (20 points)

**Building Elo rating model for NBA games.**

1. Calculate the relative Elo ratings for all the games in the nba dataset.
   Make sure to adjust the ratings of the teams at the beginning of each season.
   This link https://fivethirtyeight.com/features/how-we-calculate-nba-elo-ratings/ shows the steps for calculating elo ratings in nba. **(10 points)**

```
get_elos <- function(final_elos, def_elo = 1505){
  starting_elos <- c()
  if(length(final_elos) == 0){
    starting_elos = def_elo
  }
  else{
    starting_elos <- (def_elo * 0.25) + (final_elos * 0.75)
  }
  return(starting_elos)
}

make_final_elos_df <- function(final_elos){
```

```
  final_elos <- as.data.frame(final_elos)
  final_elos$TEAM <- rownames(final_elos)
  colnames(final_elos) <- c("ELO", "TEAM")
  rownames(final_elos) <- 1:length(final_elos$TEAM)
  final_elos <- final_elos[c("TEAM", "ELO")]
  return(final_elos)
}


nba_elos_relative <- function(data = nba2009_2018, home_factor, k_factor, def_elo = 1505){
  relative_df <- c()
  final_elos <- c()
  for(season in unique(data$SEASON_ID)){
    starting_elos <- get_elos(final_elos, def_elo)
    season_data <- data[data$SEASON_ID == season,]
    season_elos <- elo.run(score(home.PTS, away.PTS) ~
                             adjust(home.TEAM_NAME, home_factor) +
                             away.TEAM_NAME, data = season_data,
                           k = k_factor, initial.elos = starting_elos)
    season_final_elos <- final.elos(season_elos)
    season_elos <- data.frame(SEASON = season,season_elos)
    relative_df <- rbind(relative_df, season_elos)
    final_elos <- season_final_elos
  }
  final_elos <- make_final_elos_df(final_elos)
  return(list(relative = relative_df, elos_df = final_elos))
}


nba_rel_elos <- nba_elos_relative(nba2009_2021, 100, 20)
nba_rel_elos_df <- nba_rel_elos$elos_df
nba_rel_seasonal <- nba_rel_elos$relative
```

2. Interpret the predictive power of your model using either confusion matrix or brier score. **(5 points)**

```
make_conf_matrix <- function(data = nba_rel_seasonal){
  rel_seasonal_conf <- table(Actual = data$wins.A,
                             Predicted = data$Predicted)
  colnames(rel_seasonal_conf) <- c("Win", "Lose")
  rownames(rel_seasonal_conf) <- c("Win", "Lose")
  accuracy <- sum(diag(rel_seasonal_conf))/sum(rel_seasonal_conf)
  return(list(table = rel_seasonal_conf, accuracy = accuracy))
}

calculate_brier <- function(data = nba_rel_seasonal){
  diff <- data$p.A - data$wins.A
  accuracy <- sum(diff^2) / nrow(data)
  return(accuracy)
}

nba_rel_seasonal <- nba_rel_seasonal %>%
  mutate(Predicted = ifelse(p.A > 0.5,1,0))
```

```
(conf_seasonal <- confusionMatrix(data = factor(nba_rel_seasonal$wins.A),
                                  reference = factor(nba_rel_seasonal$Predicted)))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##          0 2545 3933
##          1 1391 7559
##
##                Accuracy : 0.6549
##                  95% CI : (0.6474, 0.6624)
##     No Information Rate : 0.7449
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.2511
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.6466
##             Specificity : 0.6578
##          Pos Pred Value : 0.3929
##          Neg Pred Value : 0.8446
##              Prevalence : 0.2551
##          Detection Rate : 0.1650
##    Detection Prevalence : 0.4199
##       Balanced Accuracy : 0.6522
##
##        'Positive' Class : 0
##
```

```
(calculate_brier(nba_rel_seasonal))
```

```
## [1] 0.2148968
```

```
(make_conf_matrix(nba_rel_seasonal))
```

```
## $table
##       Predicted
## Actual  Win Lose
##   Win  2545 3933
##   Lose 1391 7559
##
## $accuracy
## [1] 0.6549131
```

3. Make predictions for the same games you chose in **Problem 2.7**, but now use Elo rating model. **(3 points)**

```
get_teams_elo <- function(data = nba_rel_elos_df, team){
  return((data %>%
            filter(TEAM == team))$ELO
  )
}

games_df %>%
  rowwise() %>%
  mutate(wins.A = elo.prob(elo.A = get_teams_elo(team = home.TEAM_NAME),
                           elo.B = get_teams_elo(team = away.TEAM_NAME)),
         wins.B = 1 - wins.A)
```

```
## # A tibble: 3 x 6
## # Rowwise:
##   home.TEAM_NAME     away.TEAM_NAME       ht_w  at_w wins.A wins.B
##   <fct>              <fct>               <dbl> <dbl>  <dbl>  <dbl>
## 1 Washington Wizards Charlotte Hornets   0.514 0.486  0.353  0.647
## 2 Philadelphia 76ers Cleveland Cavaliers 0.473 0.527  0.692  0.308
## 3 Houston Rockets    Miami Heat          0.505 0.495  0.140  0.860
```

4. Compare the results **(2 points)**

The elo rating modelling approach gives completely different results, let's wait for the games and compare them.

5. Try to play with the home advantage parameter, K value or anything else, either on football data or in nba and check if that will help to improve your models' predictive power.
   **(Bonus question 10 points)**